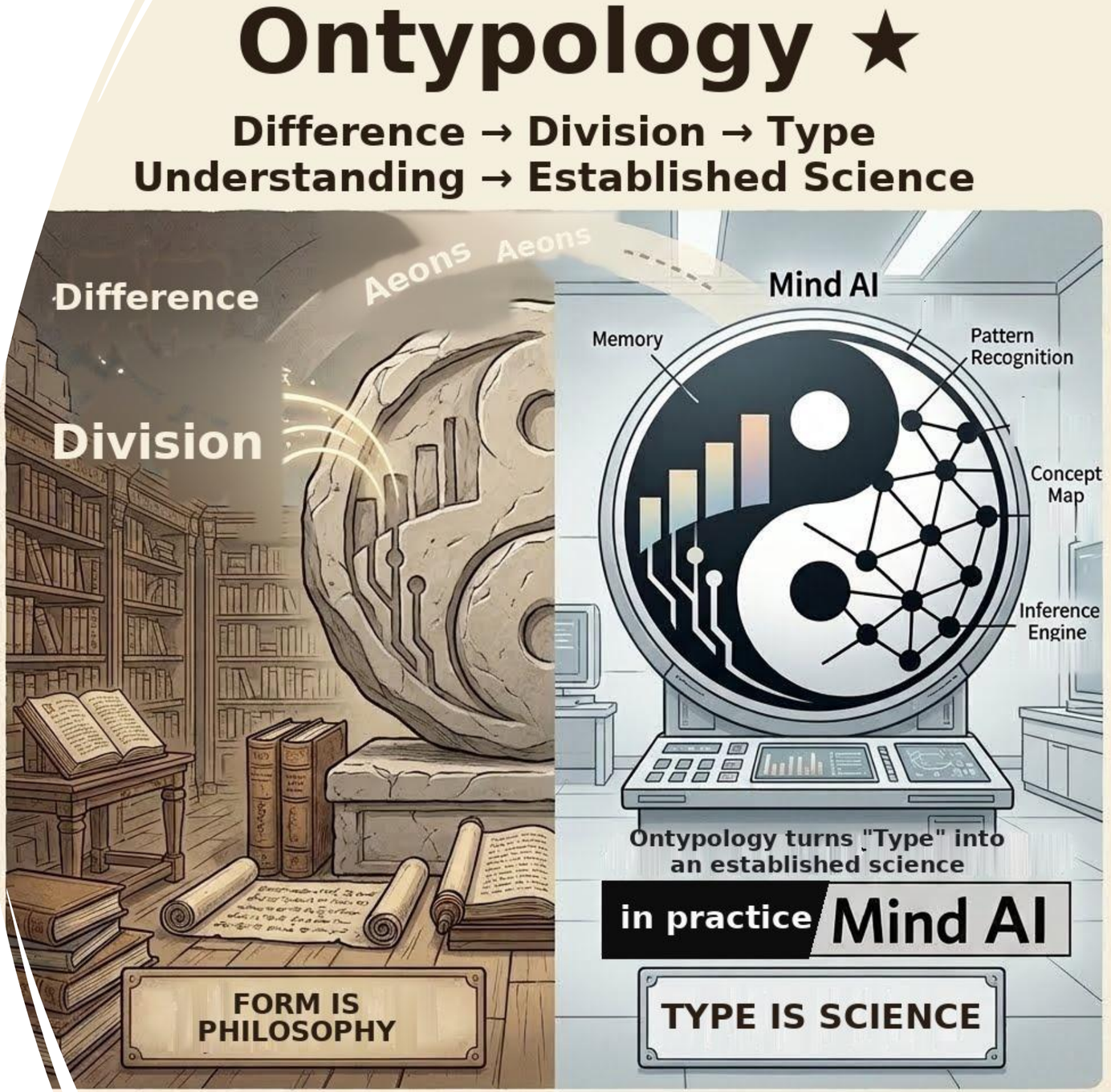
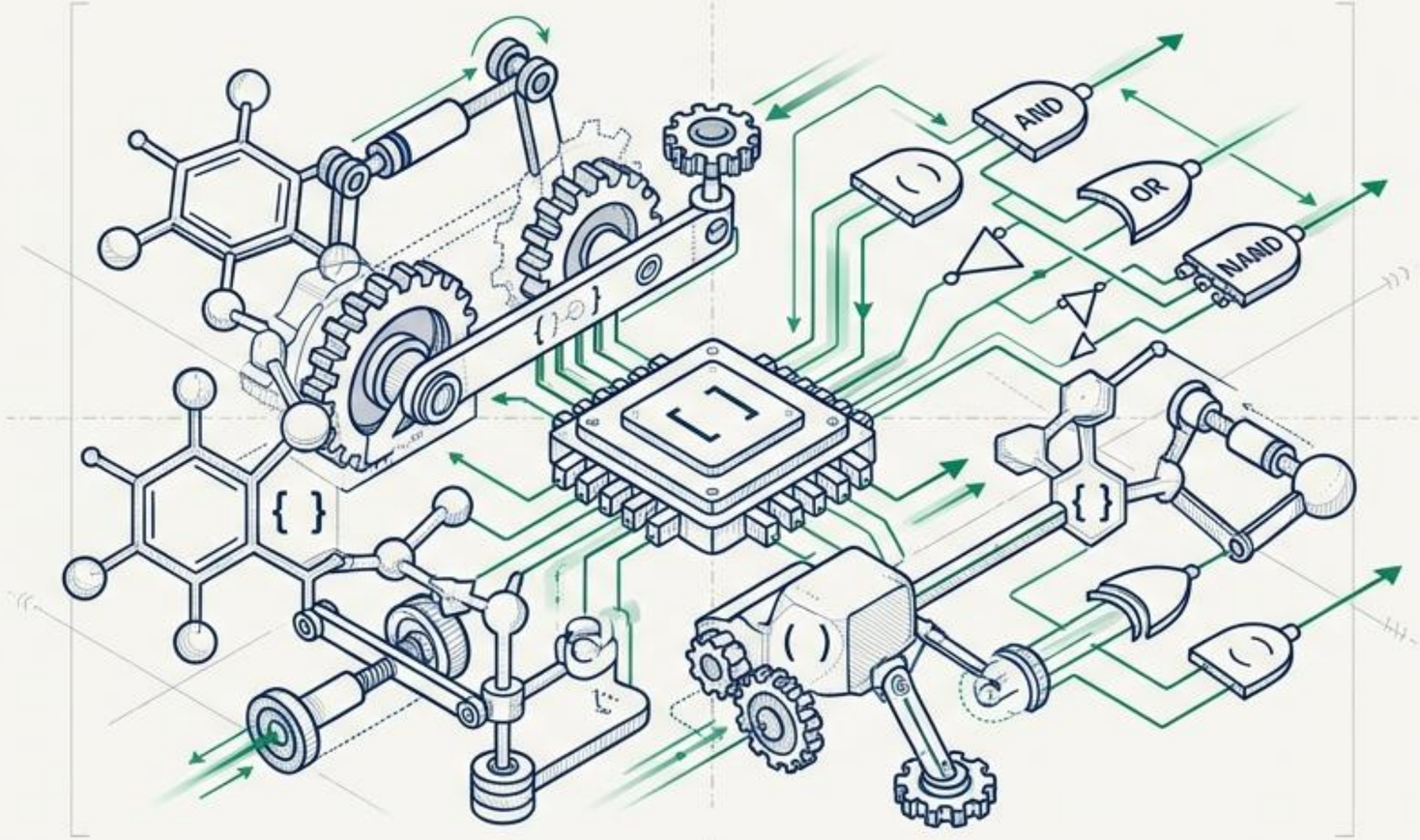
Ontypology

- From difference to division, from division to type, from type to understanding – the brain has walked this path for aeons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.





AI Drug R&D Brain: a Reasonable R&D Twin

Based on the Space Type and Time Type of Ontypology

MIND AI

Four scenes of inaccurate output

DIAGNOSTIC REPORT 01

No standing form: molecules known by nickname; rename them and memory is lost.

[FAULT: IDENTITY_LOSS]

DIAGNOSTIC REPORT 02

Derivation break: “chemical intuition” jumps to a conclusion — not reproducible.

[FAULT: LOGIC_BREAK]

DIAGNOSTIC REPORT 03

Validation drifts: only the best run is reported; raw data is missing.

[FAULT: DATA_BIAS]

DIAGNOSTIC REPORT 04

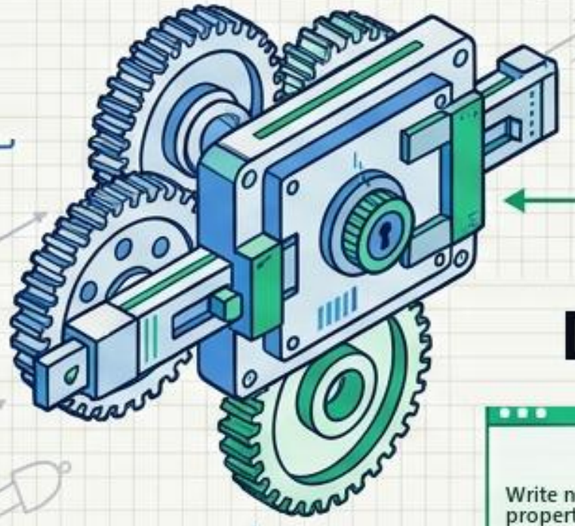
Review amnesia: the same pit is hit once every quarter.

[FAULT: RECURRING_FAILURE]

A reasoning twin: Type + Data

Type

Write structure, composition, container and criteria. Answers “what it is”.

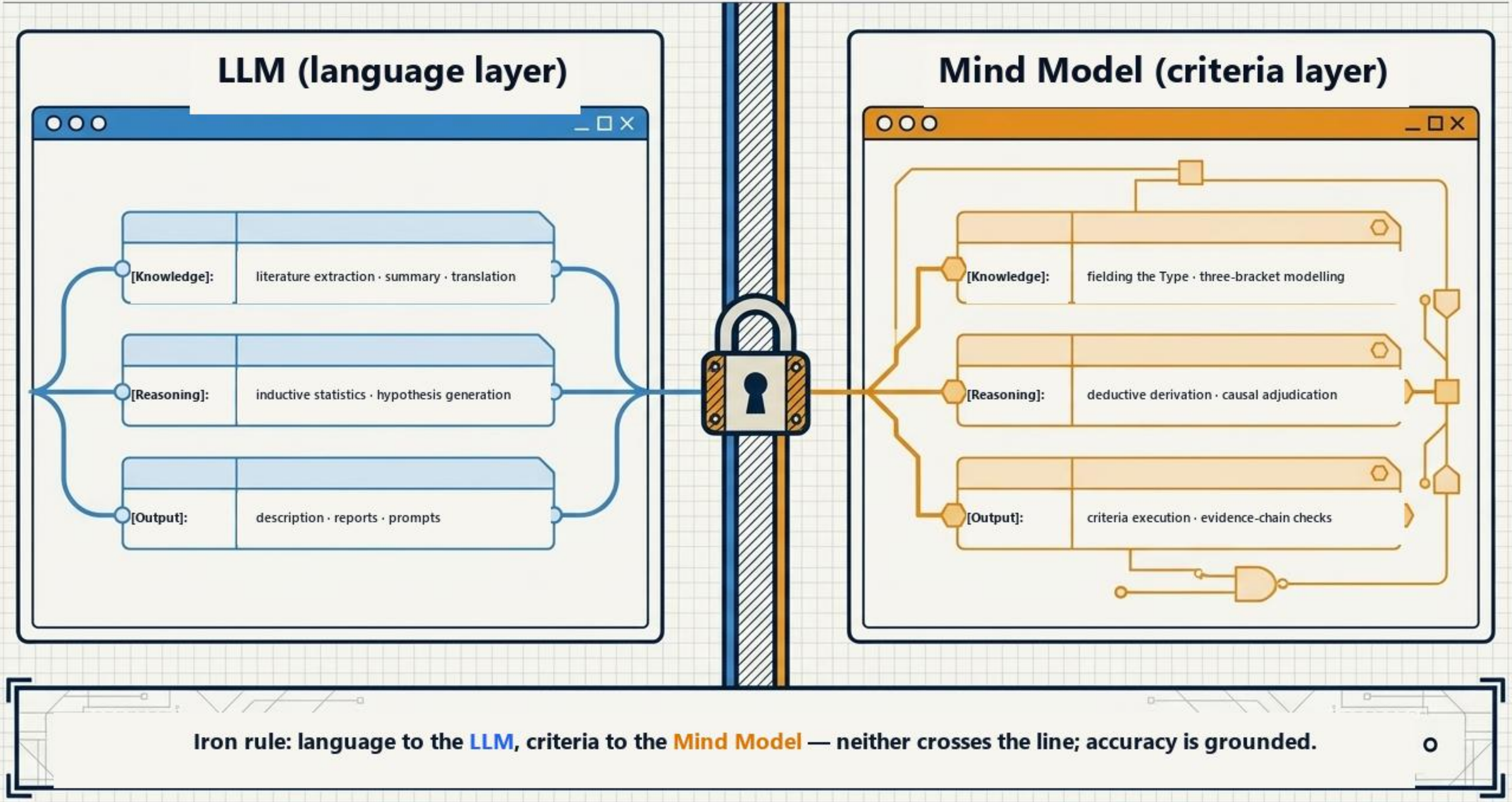


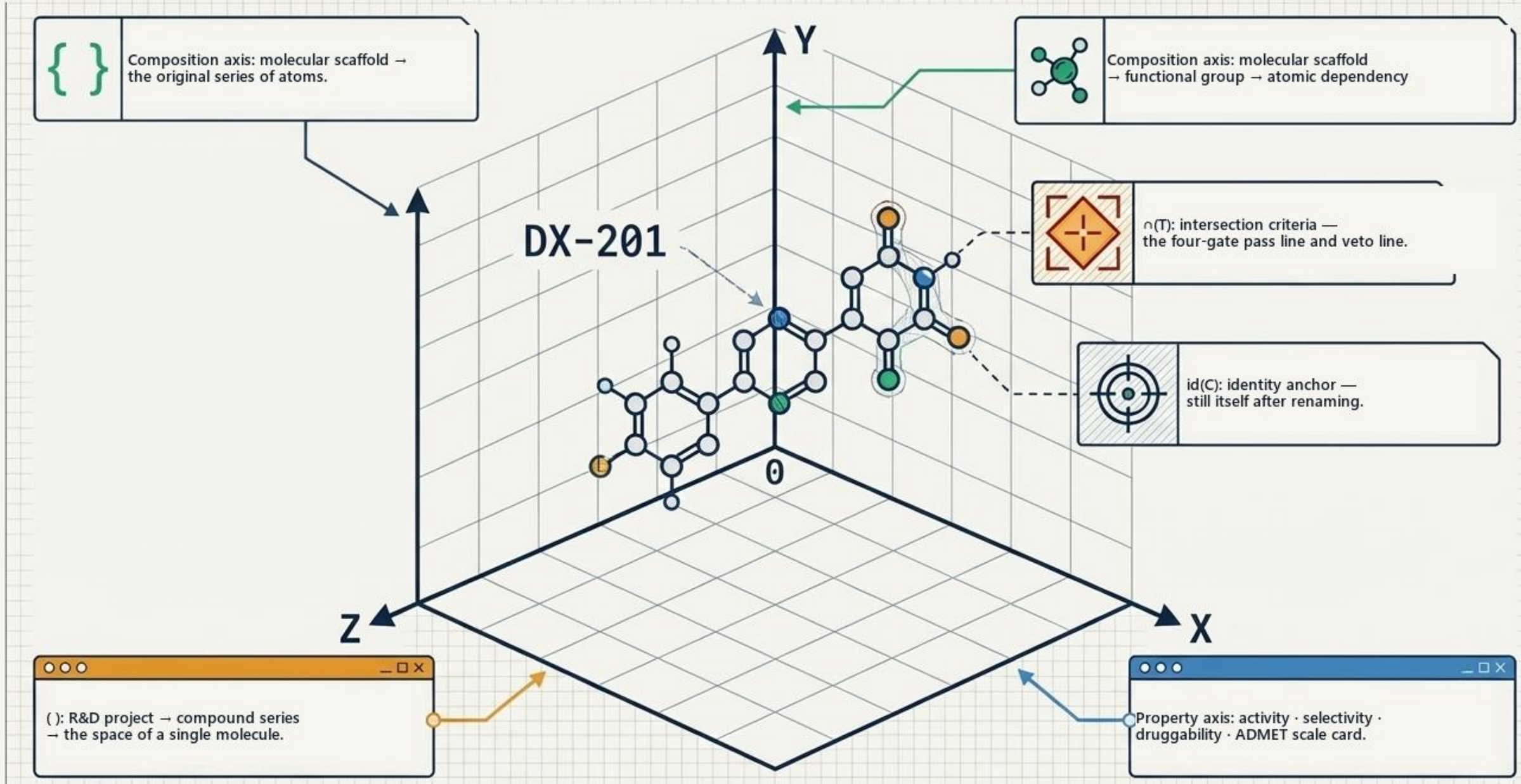
Data

Write numbers for activity, properties and relationships. Answers “how much”.

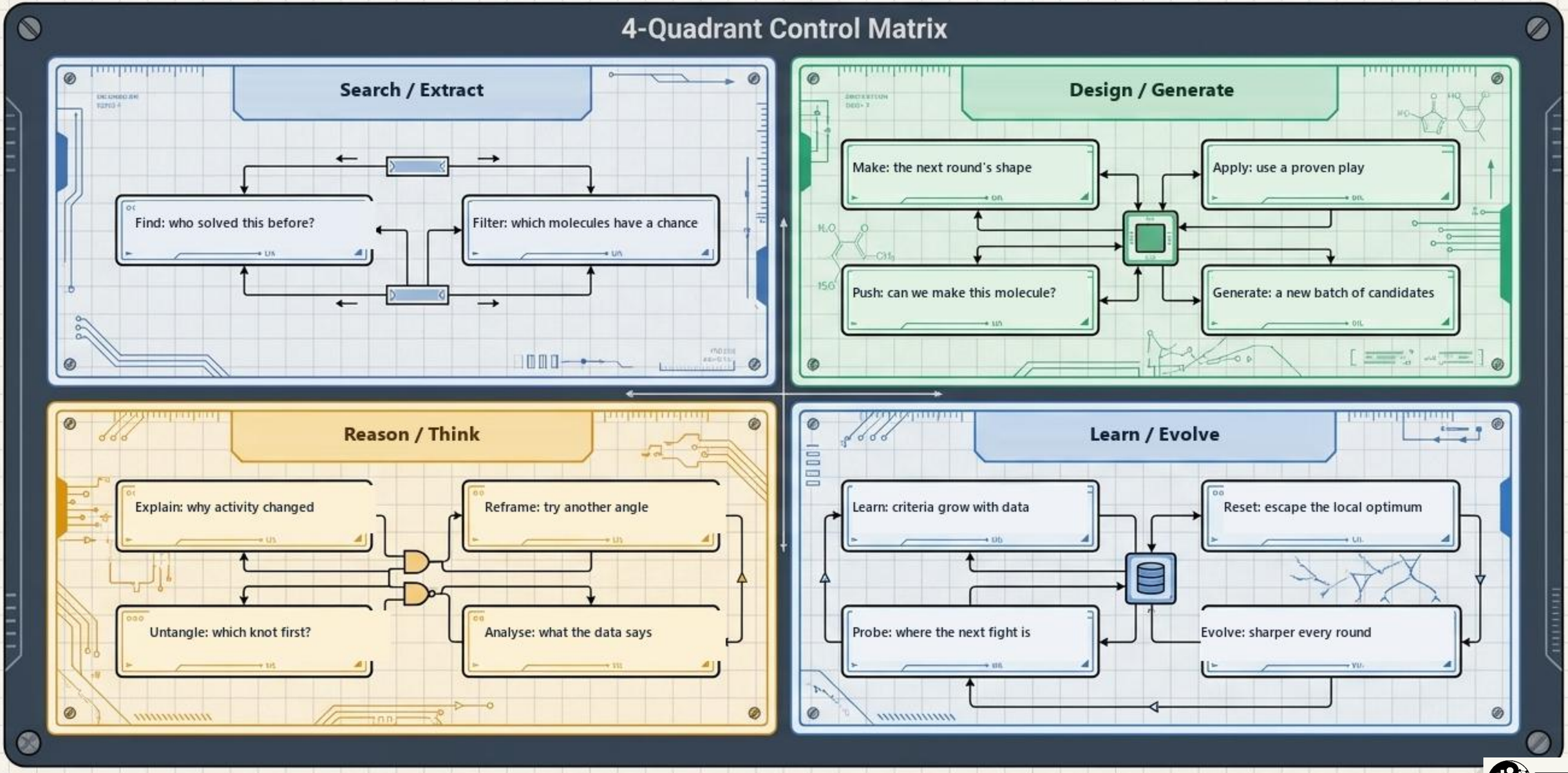
Both are indispensable. Before AI speaks, it must ask “what it is”; before concluding, it must show evidence.

MIND AI


LLM (language layer)
[Knowledge]:
literature extraction · summary · translation
[Reasoning]:
inductive statistics · hypothesis generation
[Output]:
description · reports · prompts
Mind Model (criteria layer)
[Knowledge]:
fielding the Type · three-bracket modelling
[Reasoning]:
deductive derivation · causal adjudication
[Output]:
criteria execution · evidence-chain checks
Iron rule: language to the LLM, criteria to the Mind Model — neither crosses the line; accuracy is grounded.


{ }
Composition axis: molecular scaffold → the original series of atoms.
Y
Composition axis: molecular scaffold → functional group → atomic dependency
DX-201
∩(T): intersection criteria — the four-gate pass line and veto line.
id(C): identity anchor — still itself after renaming.
0
Z
X
( ): R&D project → compound series → the space of a single molecule.
Property axis: activity · selectivity · druggability · ADMET scale card.

14 Motions of Time: the R&D verbs of the Time Type



MIND AI

14 Motions of Time: the R&D verbs of the Time Type

Discovery Phase

[Target Type]

What to attack?
Target and indication
Open a card.

[Hit Type]

What was found?
Hits and screening
Open a card.

Optimization Phase

[SAR Type]

What is the relation?
Change → consequence
Open a card.

[Optimize Type]

What to change?
Multi-parameter plan
Open a card.

Closure Phase

[Verdict Type]

Works or not?
PCC verdict
Open a card.

[Evolution Type]

What changes next round?
Open an evolution ledger

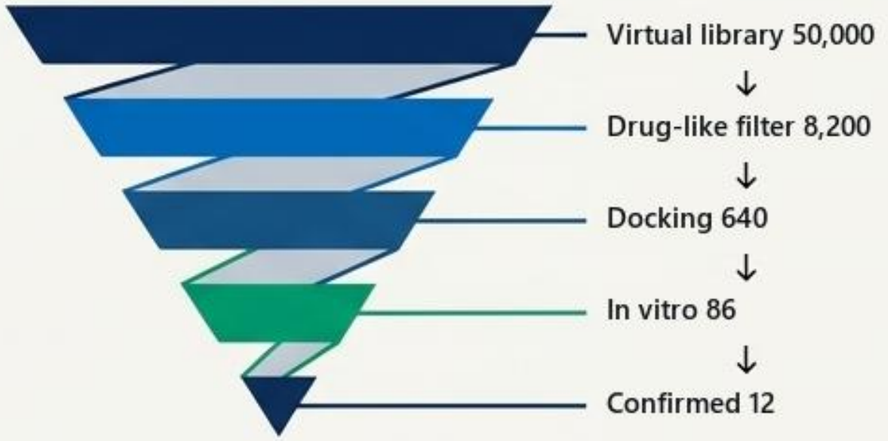
Chain rule: six forms relay; miss one and the chain breaks. Every form has a field card, criteria, and evidence.

MIND AI

Target Type field card

Field	Data
Target name	EGFR (tyrosine kinase receptor)
Target variant	T790M / C797S cis double mutation
Indication	Non-small cell lung cancer (NSCLC)
Criteria ∩(T)	Four gates (activity / selectivity / druggability / ADME), set before use
Evidence source	Timestamped and traceable.

Hit Type field card



Field	Data
Hit molecule	DX-201 (phenylpyrimidine bicyclic)
Initial activity	IC50 = 2.4 μM (n = 3, R^2 = 0.98)
Ligand efficiency	LE = 0.31 (≥ 0.3 to be listed)
Noise removal	PAINS negative, no aggregation

MIND AI

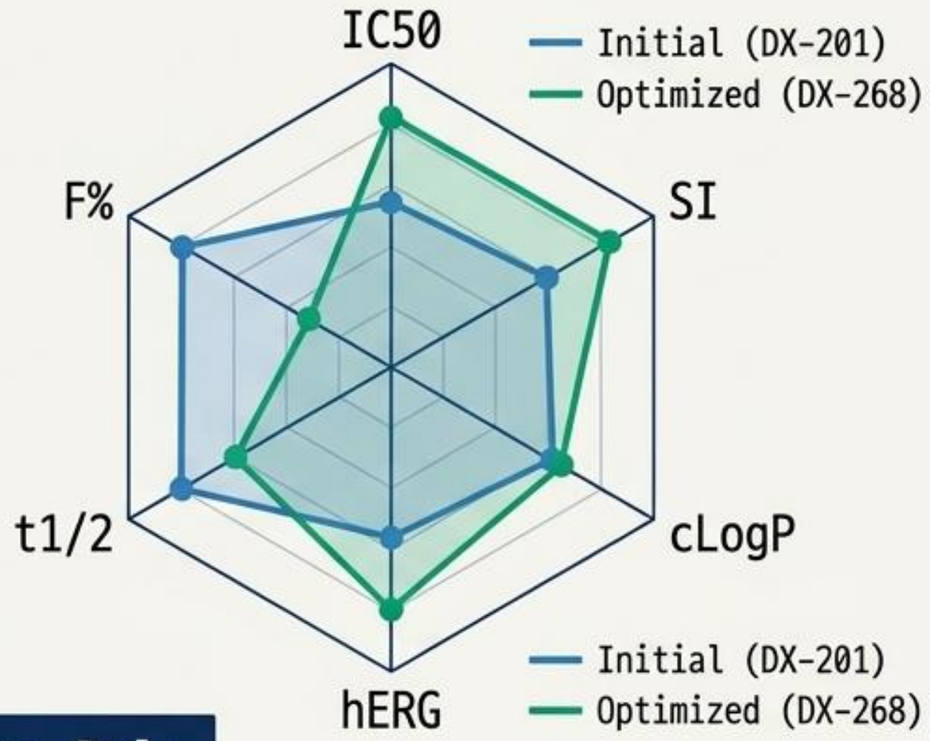
SAR Type (SAR Matrix)

	R2 = phenyl	R2 = pyrimidine
R1 = H	2.4 μM (DX-201)	
R1 = Me		
R1 = OMe		24 nM (DX-268)

Logic Rule

remove counterfactual: remove the key part (remove level 3); it must still recognise itself.
Swap back H — activity must fall back to μM.

Optimize Type (MPO Radar)



Core Rule

No axis may regress: activity gains cannot be paid for with metabolism, safety, or solubility.
Predict first, always carry a tolerance range.

MIND AI

Verdict Type (VerdictType)

[ ] Subject:

DX-288 (identity anchor id(C) aligned)

[ ] Four-Gate Funnel:

{ Activity √ | Selectivity √ |
Druglikability √ | ADME √ }

(every item backed by data; no bundling)

[ ] Conclusion:

Upgrade PCC — one of: advance / hold / shelve

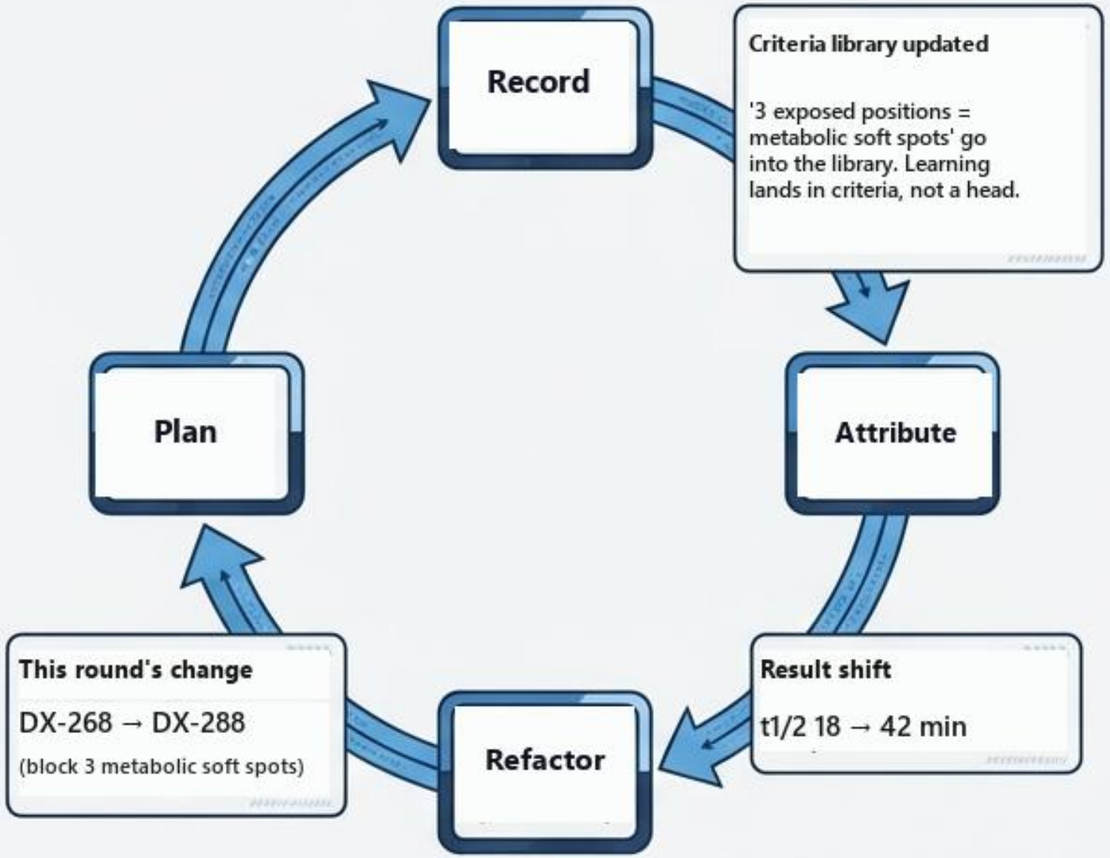
[ ] Principle:

Record pass/fail and evidence — never mood.

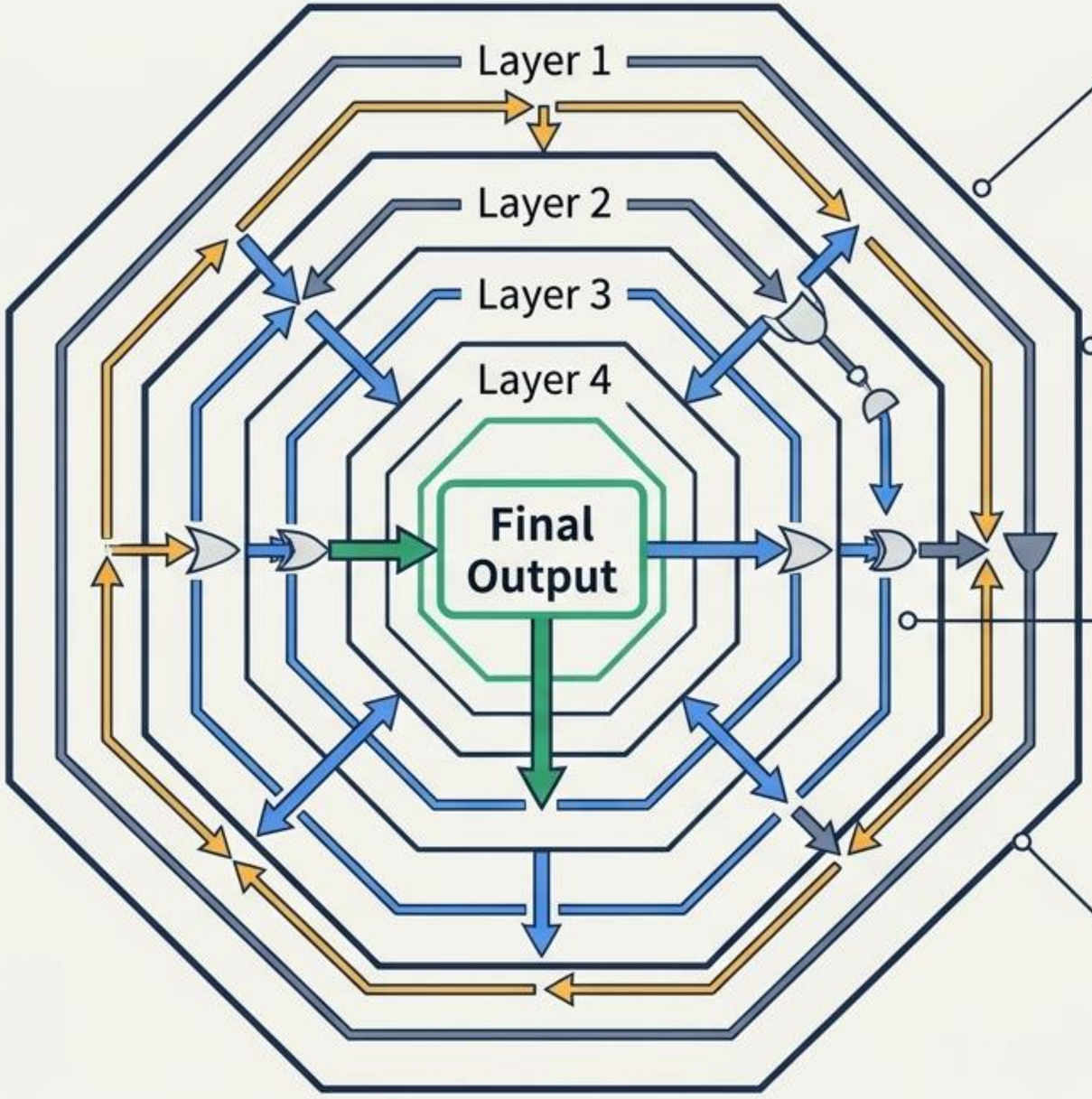
VERIFIED

DIGITAL LEDGER RECORD – PCC UPGRADE APPROVED

Evolution Type (EvolutionType)



MIND AI



Layer 1 · Eligibility gate: ∩(T)

Four-gate funnel: activity, selectivity, druggability, ADME; every change leaves a trace.

Layer 2 · Reasoning gate: deduction

Triad: premise (DX-268 t1/2 = 18 min) → rule (block soft spots to extend t1/2) → prediction (range 25–45 min). No skipping steps.

Layer 3 · Reality gate: validation

Activity, ADMET, safety window. Every set needs n ≥ 3 controls and statistics; no raw file, no trust.

Layer 4 · Audit gate: evidence chain

Form → output → derivation → validation → verdict → archive. Break one station and it rolls back.

MIND AI

Framework in four steps (read the situation)

1. 1

1. Set the container (project and stage)

2. 2

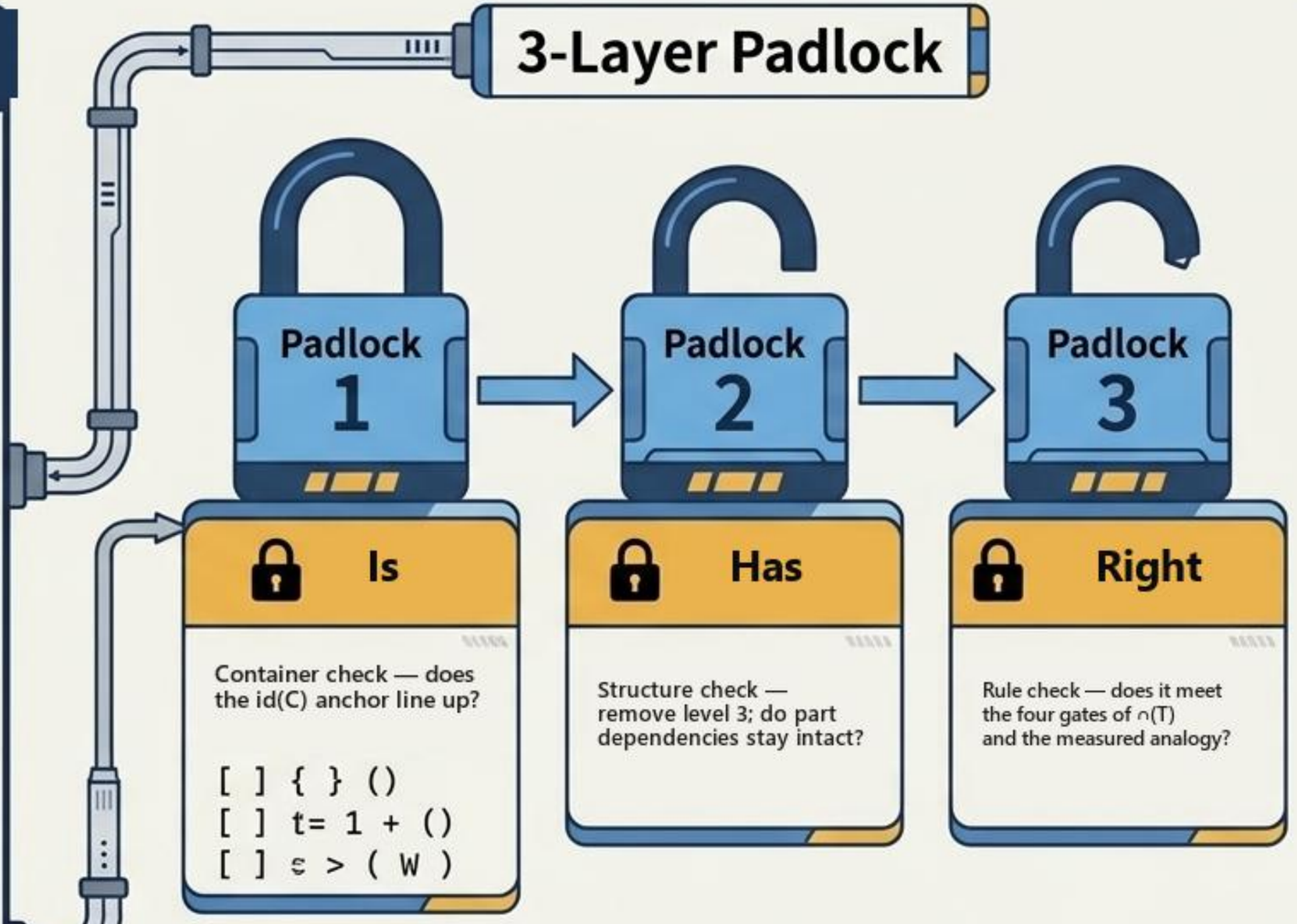
2. Carve attributes (what success looks like)

3. 3

3. Break down composition (which parts)

4. 4

4. Set criteria (how pass/fail is judged)



MIND AI

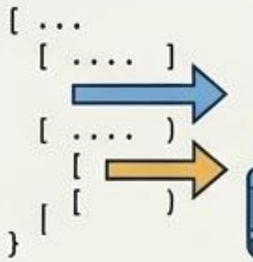
S1 Identify situation

[Input]
Project background

→

[Output]
Situation card

(fault tolerance / full success picture)





S2 Target Type

[Input] Situation card

→

[Output] Target card

(formed by four gates)





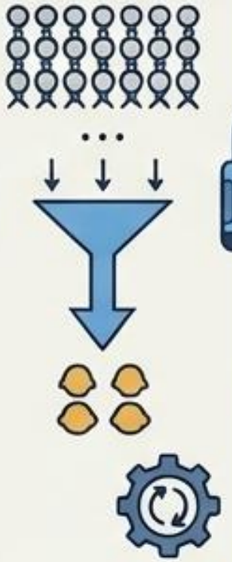
S3 Hit Type

[Input] Target Type

→

[Output] Hit card

(screening reproducible)



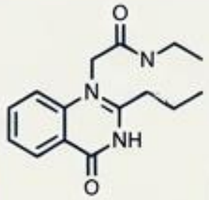
S4 SAR Type

[Input] Hit Type

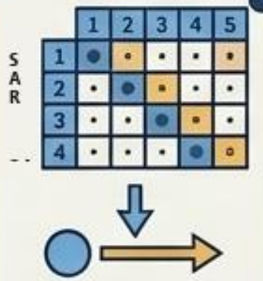
→

[Output] SAR matrix

(causal chain + counterfactual)



SAR matrix



(causal chain + counterfactual)

S5 Optimize Type

[Input] SAR Type

→

[Output]
Optimization plan

(quantifiable goals / exit criteria)





S6 Verdict Type

[Input]
Optimization plan

→

[Output] Verdict doc

(four gates checked item by item)





S7 Evolution Type

[Input] Verdict doc

→

[Output]
Evolution ledger

(attribution + executable plan)





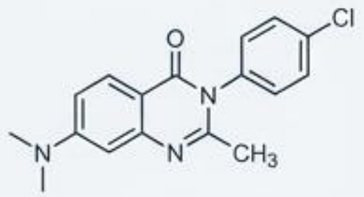
Execution Rule: If the previous stage fails, the next never starts; no number without evidence.

MIND AI

Evolurical Specification

Hit (DX-201)

Structure: phenylpyrimidine (bicyclic)

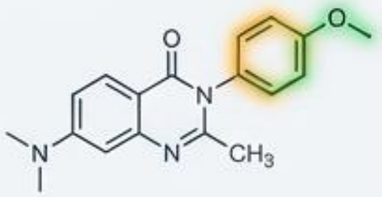


Structure:

Stats: IC_{50} = 2.4 µM

Lead established (DX-245)

Structure:

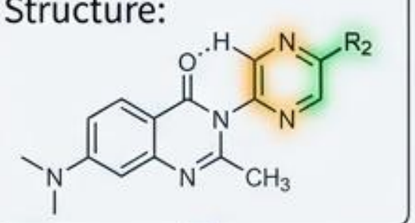


Action: R1=OMe
affinity tuning

Stats: IC_{50} = 86 nM
(SI=62, t1/2=26m)

Activity breakthrough, but ADME stuck (DX-268)

Structure:

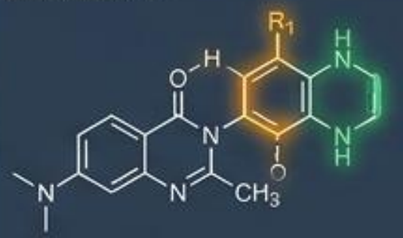


Action: R2=pyrimidine forms a second H-bond

Stats: IC_{50} = 24 nM,
but t1/2 = 18 min
(ADME gate not passed)

PCC verdict passed (DX-288)

Structure:



Action: block 3 metabolic soft spots

Stats: IC_{50} = 12 nM,
t1/2 = 42 min,
SI = 118.

PCC
(Pre-Clinical Candidate)

Synthesis Insight: Six axes met and 200× activity come from the deduction skeleton and ∩(T) funnel, round after round — not intuition.

MIND AI

Five Misuses (Traps & Solutions)

Misuses (Traps)	Solutions
Analogy as conclusion	Analogy is only a hypothesis; only ∩(T) counts.
Only the activity axis	Look at all six axes; no axis may regress.
Criteria loosened later	Change of line leaves a trace + version upgrade.
Pushing a broken chain	Broken chain: roll back and reinforce evidence.
Reviews without accounts	One page per round, mistakes never overnight.

Ten-Question Checklist

- [ ] Which container does this output belong to? Does id(C) match?
- [ ] Were the criteria set first or patched afterwards? Which version number?
- [ ] Does the prediction have an interval? Was the measurement corrected?
- [ ] Can the key conclusions be traced station by station back to the raw data?

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AI amplifies expertise; it does not replace it.

Day 1-30 (establish the Types) |

L1: six-Type field cards finalised,
four-gate criteria drafted.

Day 31-60 (establish the chain) |

L2/L3: all six evidence-chain stations
connected; two relay rounds run;
ten questions spot-checked.

Day 61-90 (establish the ledger) |

L4: evolution ledger becomes routine.
Closing: the review meeting brings
the evolution ledger, not impressions.

1. Six Types produce the output:

establish the Type before the output;
when the Type stands, accuracy follows.

2. Four-layer assurance:

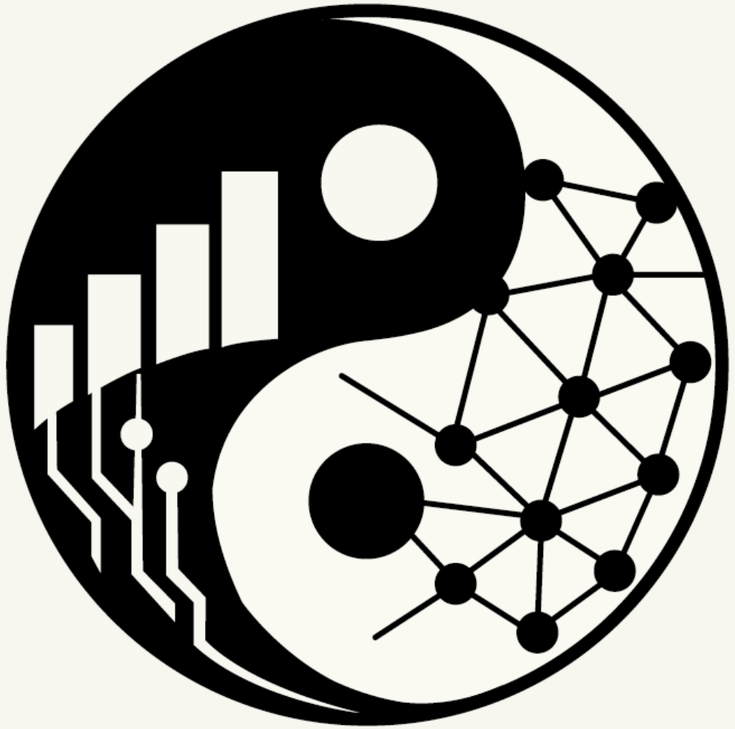
criteria · derivation · validation · chain —
all four must pass to count.

3. Evolution ledger long run:

keep accounts, mistakes become criteria;
swap moves, stagnation becomes progress.

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LonGreat Vision: to create intelligent life that understands the essence of the world



MIND AI

"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence – this is the eternal path to AGI that LonGreat pursues.

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